

INDIAN INSTITUTE OF TECHNOLOGY MANDI

SIMTiX:
A Compact Open-Source RISC-V SIMT Accelerator and
its Extension
into an End-to-End Quantized Neural-Network Inference
Engine

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by

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Declaration

I declare that the work presented in this thesis, titled “*SIMTiX: A Compact Open-Source RISC-V SIMT Accelerator and its Extension into an End-to-End Quantized Neural-Network Inference Engine*,” is my own. The register-transfer level (RTL) design, the verification testbenches, the kernels, the field-programmable gate array (FPGA) implementation flow, and the software model reported here were developed by me. Every quantitative result stated in this thesis was obtained by running the corresponding simulation or implementation flow on the design; no figure has been estimated or reproduced from another source except where it is explicitly cited or explicitly labelled as an estimate. The complete source, including the RTL, the simulation harness, and the FPGA scripts, is publicly available.

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Abstract

This thesis presents **SIMTiX**, a compact, fully open-source Single-Instruction Multiple-Thread (SIMT) accelerator for RISC-V, and its evolution from a general-purpose parallel core into an end-to-end quantized neural-network inference engine. The work is developed as a continuous engineering narrative in which every design decision is measured rather than asserted.

The baseline accelerator implements the complete SIMT mechanism set in a small, readable, synthesizable design: eight lanes organised into hardware warps, a round-robin scheduler with a background memory engine for latency hiding, address coalescing on a cache-line-wide data port, a per-warp reconvergence stack for control divergence, and an on-chip scratchpad for data reuse. It is attached to a reused five-stage RISC-V host through a memory-mapped offload interface and taken through the entire flow from RTL to a routed FPGA bitstream on a Xilinx Zynq UltraScale+ device. A central implementation result is that placing the vector register file and scratchpad in distributed RAM (LUTRAM) reduces the design from 74 % to 13 % of the device LUTs while closing timing at 100 MHz. A six-kernel benchmark suite, verified bit-exact to 1024 threads, gives a measured 7–10× speedup over the scalar host on streaming kernels and—reported honestly—a slowdown on a serial reduction, mapping precisely where the SIMT model does and does not pay.

The accelerator is then extended with a per-lane floating-point unit supporting both single (FP32) and half (FP16) precision, including the fused multiply-add and iterative divide and square root, verified bit-exact against an IEEE-754 reference over 116,089 vectors. A placed timing-closure study localises the frequency bottleneck to a single lane’s fused-multiply-add cone and, by pipelining it into three stages, meets 100 MHz and moves the critical path off the floating-point datapath; the complete system-on-chip signs off at 103.7 MHz as a deployable bitstream.

The final and principal contribution turns SIMTiX into an AI inference accelerator. A custom INT8 packed dot-product instruction, `pdot8`, is added as a pipelined DSP background engine that costs 32 additional DSP blocks and about 1 % of the LUTs while leaving the operating frequency unregressed. On top of it, quantized GEMM and GEMV kernels and an INT32-to-INT8 requantization step complete the operator set for a fully-connected network. These are composed into a trained, quantized $784 \rightarrow 128 \rightarrow 10$ multilayer perceptron that classifies MNIST handwritten digits. The network runs end-to-end on the integrated chip: the host CPU streams the weights out of an on-chip block-RAM store into the accelerator’s working memory in tiles, launches the kernels over the memory-mapped interface, and reads back the predicted digit. The chip reproduces the integer-exact

software golden on all 64 test images, bit-for-bit, demonstrating a programmable RISC-V SIMT core that performs real, trained, quantized inference on measured hardware.

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Chapter 1

Introduction

1.1 Motivation

Graphics processing units (GPUs) execute thousands of scalar threads in lockstep groups called *warps* under a Single-Instruction Multiple-Thread (SIMT) model. The microarchitectural mechanisms that make this efficient—warp scheduling for latency hiding, memory coalescing, control-divergence handling, and on-chip scratchpads—are well documented at the level of commercial GPUs, but they are hard to *study in register-transfer level (RTL) form*. Production GPUs are proprietary; open GPGPU cores such as Vortex are powerful but large and difficult to bring up; and small academic cores are often simulator-only, stopping short of a real implementation flow. There is a gap between a design that is small enough to read end-to-end and one that is complete enough to run a real workload on real hardware.

At the same time, the dominant workload for accelerators has shifted to neural network inference, and inference no longer runs in floating point. It runs in *quantized integer* arithmetic—most commonly INT8—because four INT8 values pack into one 32-bit word, quadrupling both the effective memory bandwidth and the arithmetic density of a datapath, while post-training quantization holds accuracy within a fraction of a percent of the floating-point reference. A credible accelerator today must therefore express low-precision arithmetic efficiently and demonstrate it on a real network.

This thesis addresses both gaps with a single design. It builds a compact, fully-open SIMT accelerator whose entire path—RTL, simulation, synthesis, place-and-route, and a routed bitstream—is reproducible on one workstation with free or widely available tools, and it then extends that accelerator, step by measured step, into an engine that performs end-to-end quantized neural-network inference. The design is small enough to be read as a teaching artefact yet complete enough that a trained network runs on it and produces correct answers on measured silicon-class hardware.

1.2 Objectives

The thesis pursues four concrete objectives:

1. Design a complete, synthesizable SIMT accelerator—warp scheduling, latency

hiding, memory coalescing, a reconvergence stack, and a scratchpad—and integrate it with a five-stage RISC-V host as a self-contained system-on-chip (SoC) that is taken all the way to a routed FPGA bitstream.

2. Establish, by measurement, the FPGA cost and performance of the design, including the implementation insight that a distributed-RAM register file, rather than the arithmetic units, is what makes a SIMT datapath small and fast on FPGA.
3. Extend the engine with a per-lane floating-point unit in both single and half precision, verify it bit-exact against an IEEE-754 reference, and close its timing to 100 MHz through a diagnosis-driven pipelining of the fused multiply-add.
4. Turn the accelerator into an AI inference engine by adding an INT8 dot-product instruction, quantized GEMM/GEMV and requantization kernels, and a complete on-chip demonstration of a trained, quantized MNIST classifier running end-to-end through the real chip datapath.

1.3 Contributions

The specific contributions, each backed by a measured result in this thesis, are:

- A compact, fully-open SIMT accelerator (eight lanes, four resident warps) integrated with a five-stage RISC-V host as a self-contained SoC, exercising the entire RTL-to-bitstream flow (Chapters 3–4).
- An implementation study showing that moving the vector register file and scratchpad into distributed RAM reduces LUT usage by $5.7\times$ and closes 100 MHz timing, with an analysis of why the register-file read multiplexer, not the ALUs, dominates a naive SIMT datapath (Chapter 4).
- A six-kernel benchmark suite, verified bit-exact at up to 1024 threads, that isolates distinct microarchitectural behaviours, and a measured speedup over the scalar host—including a deliberately reported case where SIMT loses (Chapter 4).
- A per-lane floating-point extension covering FP32 and FP16, in which half precision is obtained almost for free from the single-precision path by NaN-boxed widen–compute–narrow, verified bit-exact and closed to a deployable 103.7 MHz SoC by a diagnosis-driven three-stage FMA pipeline (Chapter 5).
- A custom INT8 packed dot-product instruction (`pdot8`) implemented as a pipelined DSP background engine that adds INT8 throughput without regressing frequency, and a quantized operator library (GEMM, GEMV, requantization) built on it (Chapters 6–8).

- An end-to-end, on-chip quantized MNIST inference engine in which the host CPU streams weights from block RAM into the accelerator’s working memory in tiles, and the integrated chip reproduces the integer-exact golden on all 64 test images bit-for-bit (Chapter 9).

1.4 Organisation of the thesis

Chapter 2 reviews the SIMT execution model, the RISC-V host, and INT8 quantization. Chapter 3 describes the baseline SIMTiX architecture in detail. Chapter 4 presents the FPGA implementation, the LUTRAM result, and the benchmark evaluation. Chapter 5 develops the floating-point extension and its timing-closure study. Chapter 6 introduces the `pdot8` INT8 instruction; Chapter 7 builds the quantized GEMM and GEMV kernels on it; Chapter 8 adds requantization. Chapter 9 composes these into the end-to-end on-chip MNIST inference engine—the capstone. Chapter 10 consolidates and discusses all results, and Chapter 11 concludes.

Chapter 2

Background

2.1 The SIMT execution model

A SIMT machine groups scalar *threads* into *warps* that share a single instruction stream and execute on a vector of *lanes*. SIMTiX uses a warp size equal to its lane count ($\text{WARP_SIZE} = \text{NUM_LANES} = 8$), so one warp occupies all lanes for one instruction. Several warps are resident at once ($\text{NUM_WARPS} = 4$) and time-share one fetch/decode/execute datapath; when one warp stalls on memory, the scheduler issues another, hiding the latency. Three effects determine SIMT efficiency:

- **Occupancy:** there must be enough resident warps to keep the datapath busy while others wait on memory.
- **Memory coalescing:** the lanes' individual memory accesses are merged into wide transactions where they are contiguous, so that a warp's load of adjacent addresses costs one transaction rather than eight.
- **Control divergence:** when lanes of one warp take different branch outcomes, the divergent paths must be serialized, which lowers the fraction of lanes doing useful work.

SIMTiX implements hardware for all three and instruments each so it can be measured directly.

2.2 The RISC-V host and the offload model

The accelerator is not a standalone processor; it is an *offload* engine attached to a host CPU, exactly as a discrete GPU is mapped into a host's physical address space. The host is a reused, unmodified five-stage RISC-V (RV32I) pipeline. Its only attach point is the data-memory interface: a single address decode routes accesses either to the CPU's local memory or to the accelerator's memory-mapped register page. A stock RISC-V program therefore drives the accelerator with ordinary load and store instructions—it programs a command block (kernel entry address, base pointers, thread count), writes a GO bit, and polls a DONE bit. No custom host instructions are required for control.

2.3 INT8 quantization for inference

A quantized neural network represents activations and weights as low-precision integers instead of floating-point numbers. Under *symmetric per-tensor* quantization, a real tensor x is mapped to signed 8-bit integers by a single scale factor S : $x \approx S \cdot q$, with $q \in [-127, 127]$. A layer then computes its dot products in the integer domain, accumulating products of INT8 operands into a wide INT32 accumulator that never saturates for realistic reduction lengths. To feed the *next* INT8 layer, the INT32 accumulator must be rescaled back to a signed byte—*requantization*:

$$\text{out} = \text{clamp}(\text{round}(\text{acc} \cdot \text{scale} + \text{zp}), -128, 127), \quad (2.1)$$

where $\text{scale} = (S_{\text{in}} S_{\text{weight}})/S_{\text{out}}$ composes the input, weight, and output scales, and zp is a zero point (zero for symmetric quantization). The attraction of INT8 is fourfold data density (four bytes per 32-bit word), fourfold compute density relative to FP32 for the same register pressure, and a mapping of the multiplies onto the FPGA’s hard DSP blocks—at a typical accuracy loss well below one percent. Chapters 6 through 9 build precisely this arithmetic into SIMTiX.

Chapter 3

The SIMTiX Architecture

The entire SIMT engine lives in one module, `warp_pool.sv`: the scheduler, lanes, register file, reconvergence stack, memory engine, and scratchpad are tightly co-designed and share a single decode datapath and a single data port, so they are implemented together rather than as separate blocks. This chapter describes that engine as it exists in the RTL.

3.1 System view and the offload interface

The host's data-memory signals—write-enable, `funct3`, the ALU result used as the address, and the write data—form the bus into the accelerator. A one-line address decode on the address selects between the CPU's local data memory and the accelerator's memory-mapped I/O (MMIO) page, and the read-data return is muxed back the same way. The SoC boundary also exposes the accelerator's shared-memory master ports: kernel code is fetched over an instruction port, and data arrays are accessed over a cache-line-wide data port.

Table 3.1 lists the MMIO register map. Writing `G0` produces a one-cycle launch pulse; a small dispatcher finite-state machine (`D_IDLE`→`D_LAUNCH`→`D_RUN`→`D_DONE`) hands the whole grid to the warp pool in a single launch, counts cycles until the pool retires it, and latches the runtime into the `CYCLES` register. On `G0` the accelerator launches a grid of N threads as $\lceil N/8 \rceil$ warps; each thread receives its identity and arguments through a register-seeding convention at spawn (`a0`=thread id, `a1` . . . `a3`=base pointers, `a4`= N).

3.2 The warp pool

Four warp slots are resident at once (Figure 3.1). Each owns private architectural state—program counter, active mask, register-file partition, and reconvergence stack—and all slots share one fetch/decode/ALU datapath and one data port. A round-robin scheduler issues one runnable warp per cycle while a background memory engine advances one coalesced transaction per cycle, so one warp's memory latency is overlapped with other warps' compute.

Table 3.1: Memory-mapped command/status register map. Base address `0x8000_0000`, one 4 KB page.

Offset	Name	R/W	Description
<code>0x00</code>	<code>KERNEL_PC</code>	W	byte address of the kernel’s first instruction
<code>0x04</code>	<code>BASE_A</code>	W	base pointer, argument 0
<code>0x08</code>	<code>BASE_B</code>	W	base pointer, argument 1
<code>0x0C</code>	<code>BASE_C</code>	W	base pointer, argument 2
<code>0x10</code>	<code>N</code>	W	total thread count (grid size) to launch
<code>0x14</code>	<code>CTRL</code>	W	bit 0 = GO (write 1 to launch)
<code>0x18</code>	<code>STATUS</code>	R	bit 0 = DONE, bit 1 = BUSY
<code>0x1C</code>	<code>CYCLES</code>	R	cycle count of the last kernel

3.2.1 Scheduling and spawning

Two combinational selectors run each cycle. The *issue* selector scans from a rotating pointer for the first runnable slot and issues it, then advances the pointer (round-robin). The *fill* selector scans for the first empty or retired slot and, if grid threads remain to be spawned, drops a new warp into it: its base thread id, a fresh base stack frame, and its active mask (a tail mask handles a partial final warp when N is not a multiple of eight). Slots are recycled the instant a warp retires, so grids with far more warps than the four hardware slots run correctly—this is exactly the mechanism that later lets a single launch of $N = 128$ threads run the MNIST layers.

3.2.2 Fetch, decode, and the live top-of-stack view

The top-of-stack (TOS) frame of the issuing warp gives the cycle’s live view: its current PC drives the shared instruction fetch, and its active mask says which lanes execute. Decode is plain combinational RV32I—opcode, register fields, `funct3`, the immediate forms, and the load/store/ecall flags—with the 4-bit ALU control reused from the host core and extended for the RV32M mul. When a pushed stack frame’s PC reaches its recorded reconvergence PC, that issue slot pops the frame instead of executing, which is how divergent lane groups rejoin.

3.2.3 Per-lane datapath

For the issuing warp, all eight lanes evaluate in parallel each cycle. Per lane: the two operands are read from the register file; the ALU inputs are selected by opcode; the ALU result is computed (and doubles as the effective address for memory operations); and a writeback value and enable are formed. Writeback is gated off for a destination of `x0` and for lanes not in the active mask—masked-off lanes make no architectural change. Branch decisions are per lane, producing a taken mask; uniform decisions such as a jump target use the lowest active lane as the leader.

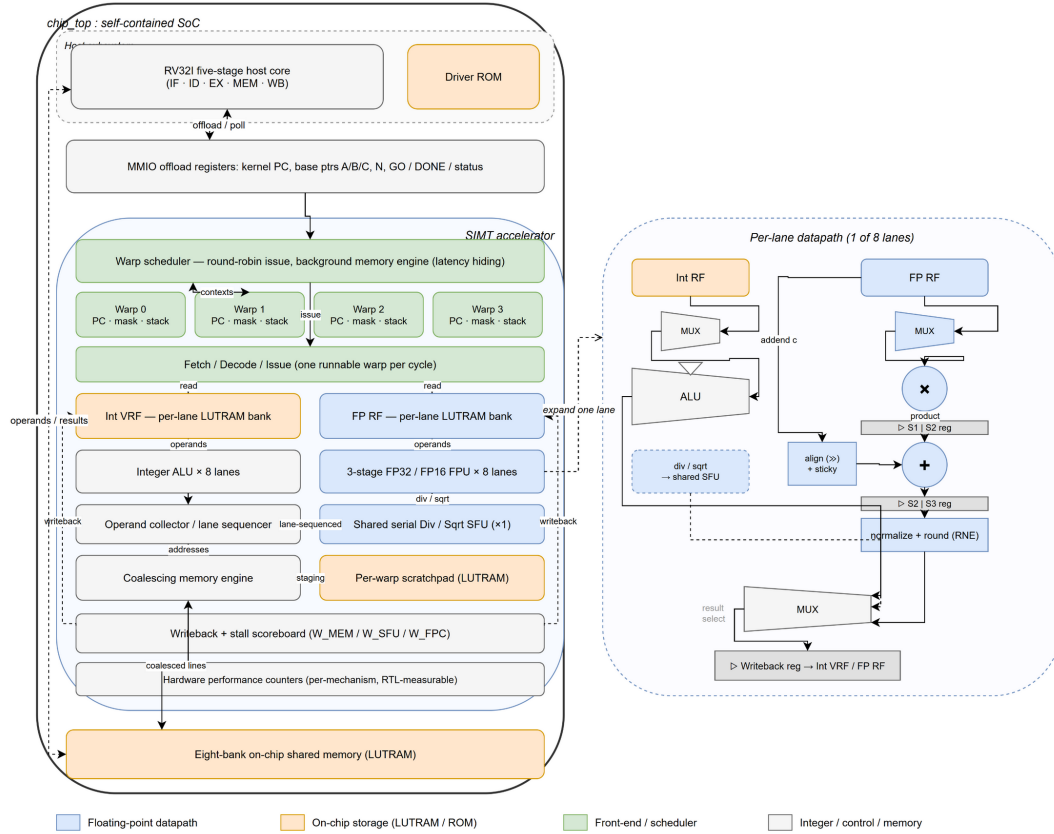


Figure 3.1: The SIMTiX accelerator datapath: the round-robin warp scheduler drives a shared fetch/decode over the top-of-stack view of the issuing warp; eight lanes execute in lockstep against a per-lane distributed-RAM register file; and a background memory engine services one coalesced line transaction per cycle while the scheduler issues other warps.

3.2.4 Control divergence: the reconvergence stack

Each warp owns a reconvergence stack of depth eight. A branch is resolved against the active mask. If all active lanes agree, the PC is simply retargeted—no divergence. If they disagree, the warp records the union mask at the join and pushes a “continue” frame so the two lane groups run in turn:

- a *forward* branch (a single-sided *if*) reconverges at the branch target, and the pushed frame runs the fall-through lanes first;
- a *backward* branch (a loop) reconverges at the fall-through (the loop exit), and the pushed frame runs the still-looping lanes.

The pushed frame pops when its PC reaches the recorded join, and the warp reconverges into the frame below, which already holds the union mask. The base frame never pops. Nesting works to the stack depth. The convention is compiler-free: general two-sided *if/else* is written as two single-sided *ifs*. This mechanism is what keeps a data-dependent kernel such as the Collatz step-count correct while lanes finish at different times.

3.2.5 The vector register file in distributed RAM

The vector register file (VRF) is logically $\text{NUM_WARPS} \times \text{NUM_LANES} \times 32$ 32-bit words. It is built as eight independent per-lane banks, each a distributed-RAM array with one synchronous write port and two asynchronous read ports, addressed by {warp, register}. Two mechanisms preserve the original flip-flop semantics exactly while keeping a single write port. First, *seed-on-read*: a per-(warp, lane, register) valid bit records whether a register has been written this grid; an unwritten register returns its spawn seed (a_0 =thread id, $a_1 \dots a_4$ =arguments, else zero), which reproduces the old “zero the file on spawn, then seed the arguments” behaviour with no clearing cycle. Second, an *arbitrated single write port*: the two producers that can target the file in one cycle—the issue stage’s compute writeback and the memory engine’s load writeback—are arbitrated so that at most one writes each lane per cycle, with the memory writeback taking priority and the compute writeback deferring (and harmlessly re-issuing) on a collision. As Chapter 4 shows, this single design decision is what makes SIMTiX small and fast on FPGA.

3.2.6 The background memory engine and coalescing

At most one memory instruction is in flight, because there is a single data port. When an issuing warp executes a load or store and the engine is free, the access context is *latched*—each lane’s effective address and store data, the active mask, and the resume PC—and the warp moves to a memory-wait state. Latching is essential because the live decode keeps tracking the currently-issued warp, not the one being replayed. The data port is a whole 256-bit line (eight words). Each cycle the engine picks the line of the lowest still-pending lane, gathers every pending lane that shares that line into one transaction, extracts or merges each lane’s word, and clears those lanes; when the mask drains it writes the resume PC back and returns the warp to the run state. The cost is the number of distinct lines touched: a contiguous warp access collapses to one transaction, a fully scattered one costs up to eight. A debug counter records line transactions, making coalescing efficiency directly measurable.

3.2.7 The scratchpad

A per-warp scratchpad of 64 words occupies a dedicated address aperture; accesses that fall in the aperture are serviced from on-chip RAM and never reach the global data port. Data reused across a warp’s lanes—for example a broadcast matrix row—can be staged once and reused, cutting global traffic. The scratchpad is shared across a warp’s lanes and held in a single distributed-RAM array with one read and one write port, so the engine serializes one lane per cycle; the semantics remain exact because within one SIMT memory instruction all lanes perform the same operation.

Table 3.2: Principal design parameters (defaults).

Parameter	Default	Meaning
XLEN	32	data width (RV32)
NUM_LANES	8	physical SIMT lanes
WARP_SIZE	8	threads per warp
NUM_WARPS	4	resident warp slots (power of two)
SDEPTH	8	reconvergence-stack depth
LINE_WORDS	8	words per coalesced line (256-bit)
SCRATCH_WORDS	64	scratchpad words per warp
MMIO_BASE	0x8000_0000	command-register aperture
SCRATCH_BASE	0x4000_0000	scratchpad aperture

3.2.8 Divergence-aware lane clock-gating

The active mask is exactly the set of lanes doing useful work; masked-off lanes produce no architectural change yet, in a baseline design, still clock their pipeline registers. Treating the active mask as a per-lane clock-enable, the lane-datapath dynamic energy that a gated design saves over a kernel is

$$E_{\text{saved}} = 1 - \frac{\sum_i a_i}{L \cdot \sum_i 1} = 1 - \frac{\text{active-lane instructions}}{L \cdot \text{issued instructions}}, \quad (3.1)$$

where an issued instruction activates a of L lanes. Two hardware counters—issued instructions and summed active lanes—make Eq. (3.1) directly observable; Chapter 4 reports the saving as a function of divergence.

3.3 Parameters

Table 3.2 lists the design parameters. All are set at elaboration, so the lane count and resident-warp count can be swept, a fact the design-space explorations of Chapters 4 and 5 exploit.

Chapter 4

FPGA Implementation and Baseline Evaluation

SIMTiX is written in synthesizable SystemVerilog and integrated with the reused five-stage RV32I host into a self-contained chip, `chip_top`: host core, driver ROM, accelerator, and an eight-bank on-chip shared memory. In the integrated chip the host loads the operands into shared memory, launches the accelerator, reads the results back, and publishes a checksum—so the whole CPU→memory→accelerator→memory→CPU loop runs in hardware. All FPGA numbers in this thesis target the Xilinx Zynq UltraScale+ `xczu7ev` (230,400 LUTs, 460,800 flip-flops, 1,728 DSP48E2 blocks).

4.1 From flip-flops to distributed RAM

The first synthesis of the accelerator with a flip-flop register file consumed 170,868 LUTs—74 % of the device—at only 10 % of its flip-flops, and topped out near 122 MHz. The cause is structural rather than arithmetic. A flip-flop VRF of $4 \times 8 \times 32$ 32-bit words is 32,768 registers read as `vrf[warp][lane][reg]` with *both* the warp and the register index variable, i.e. a 128-to-1 multiplexer per lane per read port. That multiplexer fabric, not the ALUs, dominates the area and sets the critical path. Re-expressing the VRF as one distributed-RAM bank per lane (Section 3.2.5), and likewise moving the scratchpad into LUTRAM, removes essentially all of those registers. Table 4.1 quantifies the effect: the LUTs fall by $5.7\times$ (74 % to 13 % of the device) and the design closes 100 MHz timing with positive slack. This is the single most important implementation result of the baseline: the register file, not the compute, is the cost of a SIMT datapath on FPGA, and distributed RAM is the lever.

4.2 Full-chip place-and-route

The complete chip was taken through optimisation, placement, physical optimisation, routing, and bitstream generation on the `xczu7ev`. Post-route it meets all timing at 100 MHz (setup worst negative slack +0.000 ns, hold met) and uses 34,850 LUTs (15.1 %), 6,130 flip-flops (1.3 %), 3,348 LUTRAM, 24 DSP, and zero BRAM, at 1.09 W. The

Table 4.1: Accelerator PPA on xc7z020 before and after moving the vector register file and scratchpad to distributed RAM (out-of-context synthesis).

Metric	FF-based VRF	LUTRAM VRF + scratch	
	baseline	after	Δ
LUTs	170,868 (74%)	30,088 (13%)	-82%
Flip-flops	44,112 (10%)	4,590 (1.0%)	-90%
DSP48E2	24	24	–
BRAM	0	0	–
Timing (100 MHz)	WNS -3.21 ns	WNS +2.25 ns	met
Power	1.67 W @ 200 MHz	0.81 W @ 100 MHz	–

Table 4.2: Benchmark kernels and the mechanism each one stresses.

Kernel	Computation	Stresses
vadd	$C_i = A_i + B_i$	baseline streaming
saxpy	$C_i = 3A_i + B_i$	coalescing, mul
fir	$C_i = A_i + 2A_{i+1} + A_{i+2}$	stencil, partial coalescing
relu	$C_i = \max(0, A_i)$	data-dependent light divergence
collatz	Collatz step count	data-dependent heavy divergence
reduce	$C_w = \sum_{l=0}^7 A_{8w+l}$	scratchpad + reduction

accelerator is about 87 % of the logic; the host core and the 4 KB shared memory are small by comparison. This integrated, routed chip is the bring-up vehicle whose “result = 964” checksum demonstration recurs throughout the thesis, and it is the same top-level into which the MNIST inference engine of Chapter 9 is later configured.

4.3 Verification methodology

Correctness is established at several levels: cycle-accurate RTL simulation with self-checking testbenches under Verilator; behavioural simulation in the vendor GUI; full place-and-route with post-route timing sign-off; and post-implementation *gate-level* functional simulation of the routed netlist, which reproduces the integrated chip’s golden checksum bit-for-bit. The benchmark suite below adds a breadth-oriented level.

4.4 Benchmark suite and methodology

Six kernels were written, each chosen to stress a distinct mechanism (Table 4.2). All are assembled from RISC-V source with a stock toolchain, each runs at $N \in \{8, 64, 256, 1024\}$ threads, and every output element is checked against a golden model in the testbench. For an apples-to-apples speedup, each kernel is also written as a natural scalar loop and executed on the *same* five-stage host in RTL, counting cycles from start to completion.

Table 4.3: Measured cycles and speedup at the largest size (`collatz` at $N=256$). Lane utilization and global traffic are read from the accelerator’s hardware counters.

Kernel	N	Accel	CPU	Speedup	Lane-util
<code>vadd</code>	1024	1,413	13,322	9.43×	100%
<code>saxpy</code>	1024	1,669	16,394	9.82×	100%
<code>fir</code>	1024	2,053	14,345	6.99×	100%
<code>relu</code>	1024	1,284	12,809	9.98×	93.7%
<code>collatz</code>	256	9,637	26,601	2.76×	68.3%
<code>reduce</code>	1024	6,023	4,363	0.72×	39.8%

4.5 Measured speedup, coalescing, and divergence

Table 4.3 gives the headline evaluation of the baseline: measured accelerator cycles, scalar-host cycles, the resulting speedup, and lane utilization. The results map the SIMT model’s operating envelope precisely.

Streaming kernels (`vadd`, `saxpy`, `relu`) reach 9.4–10×: they approach the eight-lane width scaled by the coalescing traffic reduction, and their lane utilization is at or near 100%. *The stencil fir* is lower at 6.99× because its shifted reads A_i, A_{i+1}, A_{i+2} straddle line boundaries and roughly double its memory traffic—the signature of *partial* coalescing, visible directly in the line-transaction counter. *The divergence-heavy collatz* achieves 2.76×: its per-lane trip count is data-dependent, so finished lanes mask off while others continue, dropping utilization to 68.3%; SIMT still wins, but the divergence erodes the advantage. *The serial reduction reduce* falls below break-even at 0.72×—it is *slower* than the scalar core. This case is reported deliberately: its second phase collapses eight lanes’ partial sums onto a single lane (utilization 39.8%) and its scratchpad is single-ported, so a reduction needs a dedicated cross-lane primitive to win on this engine. Honest reporting of where the model loses is a result in its own right.

Per Eq. (3.1), the complement of lane utilization is the lane-datapath dynamic energy a divergence-aware clock-gated design saves: from 0% on the convergent streaming kernels up to 22.5% on a divergence-heavy microbenchmark. Figures 4.1 and 4.2 plot the throughput ramp and the measured speedups.

4.6 Design-space exploration: lanes and warps

Because the lane count and resident-warp count are elaboration parameters, all twelve configurations of $\text{NUM_LANES} \in \{4, 8, 16, 32\}$ against $\text{NUM_WARPS} \in \{2, 4, 8\}$ were swept; all elaborate and pass the suite bit-exact. Table 4.4 reports the geometric-mean throughput over the five lane-count-agnostic kernels. Throughput scales almost linearly with lanes ($3.06 \rightarrow 5.85 \rightarrow 11.68 \rightarrow 23.24$ work-items/cycle, about 1.97× per doubling), and widening the lanes also widens the coalescing line, so global traffic falls

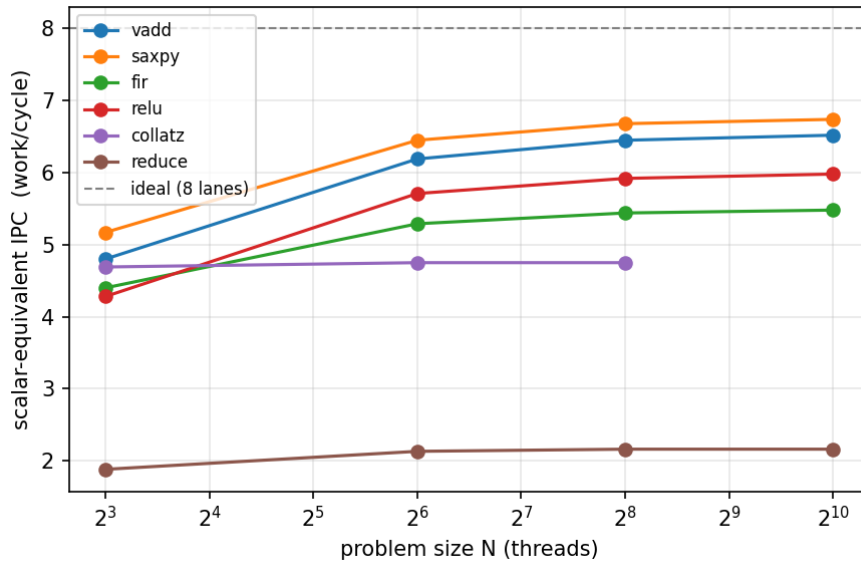


Figure 4.1: Scalar-equivalent throughput versus problem size. Throughput ramps as occupancy rises (latency hiding) and saturates below the eight-lane ideal because of the single shared data port.

Table 4.4: Geometric-mean throughput (work-items/cycle) over the five lane-agnostic kernels, swept over lanes and resident warps. All twelve configurations verify bit-exact.

NUM_LANES	NUM_WARPS		
	2	4	8
4	3.06	3.06	3.05
8	5.85	5.85	5.85
16	11.68	11.68	11.65
32	23.24	23.24	23.14

proportionally. The resident-warp count makes essentially no difference here, because the behavioural memory model answers in a single cycle and so offers almost no latency for extra warps to hide; the two knobs are separable—lanes buy data-parallel throughput, warps buy latency tolerance—and both cost register-file area.

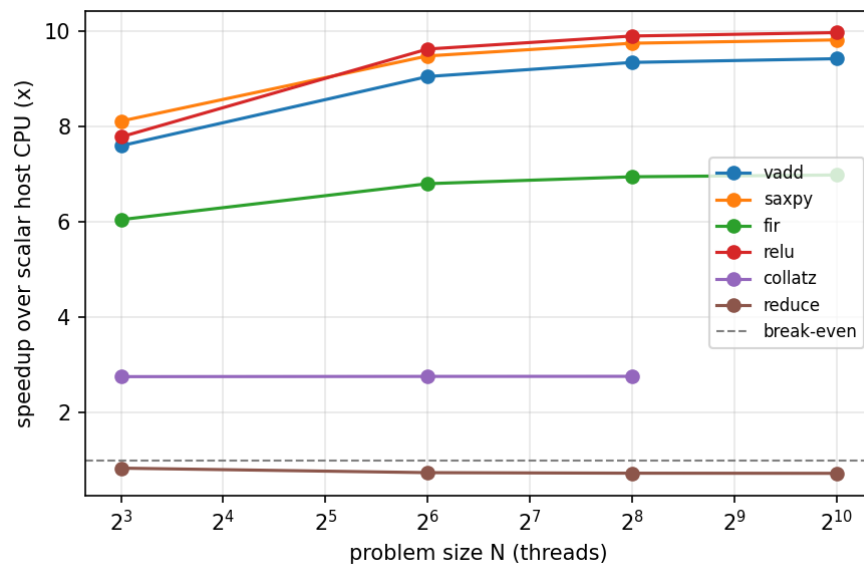


Figure 4.2: Measured speedup over the five-stage RV32I host running the same work. Streaming kernels reach 7–10×; divergence reduces the gain; the serial reduction falls below break-even.

Chapter 5

Floating-Point Extension

The base engine is integer-only. This chapter extends it with a per-lane floating-point unit in both single (FP32) and half (FP16) precision, keeping the same SIMT structure: the extension is a per-lane execute unit and a second per-lane register file, not a new pipeline. It covers the full common arithmetic set—add, subtract, multiply, the fused multiply-add, and division and square root—plus the sign, compare, convert, and move operations, all in both precisions. It matters to this thesis for two reasons: it establishes the timing methodology later reused for the INT8 engine, and its rounding-correct FP path is directly reused by the requantization step of Chapter 8.

5.1 The per-lane FP unit and register file

Each lane gains a compact, fixed-latency floating-point unit beside its integer ALU, and a second per-lane register file holds the 32 floating-point registers in distributed RAM, addressed exactly like the integer VRF. The two files have independent write ports, so an FP compute result and an integer load can retire in the same cycle; only same-file collisions defer. Rounding is round-to-nearest-even, float-to-integer truncates, and subnormals are flushed to zero—a deliberate throughput-oriented choice that keeps each lane small.

5.2 Half precision by widen–compute–narrow

Half precision is not a second arithmetic datapath. A 16-bit half lives in a 32-bit register NaN-boxed (the upper 16 bits all ones), per the RISC-V convention. For an FP16 operation the unit checks the box, widens the operand exactly to FP32, runs the single-precision datapath, and rounds the FP32 result back to FP16. This is correct, not approximate: because the FP32 intermediate carries a 24-bit significand and $24 \geq 2 \cdot 11 + 2$, rounding first to FP32 and then to FP16 gives the same value as rounding directly to FP16, so FP16 add, subtract, multiply, and the integer/FP16 conversions are correctly single-rounded. Half precision thus reuses the verified single-precision logic almost entirely, adding only the widen, narrow, and box steps—and, as shown below, at no extra cycle cost.

Table 5.1: Floating-point operation coverage. Every operation is implemented and checked in both FP32 and FP16, bit-exact against an IEEE-754 reference.

Category	Operations (each in FP32 <i>and</i> FP16)
Arithmetic	<code>fadd</code> , <code>fsub</code> , <code>fmul</code>
Fused	<code>fmadd</code> , <code>fmsub</code> , <code>fnmsub</code> , <code>fnmadd</code>
Divide / root	<code>fdiv</code> , <code>fsqrt</code> (iterative, multi-cycle)
Sign / select	<code>fsgnj{,n,x}</code> , <code>fmin</code> , <code>fmax</code>
Compare	<code>feq</code> , <code>flt</code> , <code>fle</code>
Convert	<code>float↔int</code> , <code>FP32↔FP16</code>
Move / classify	<code>fmv.x._</code> , <code>fmv._.x</code> , <code>fclass</code>
Bit-exact checks vs. IEEE reference: 116,089 (0 mismatches)	

5.3 The fused multiply-add and the iterative SFU

The fused multiply-add computes $\pm(a \cdot b) \pm c$ with a *single* rounding, so it cannot be built by chaining the multiplier and adder. The unit forms the exact 48-bit product, anchors it in a wide fixed-point accumulator, aligns the addend while capturing a sticky bit, and normalizes and rounds once. Division and square root are produced one bit per cycle by a digit-recurrence and are the engine’s first multi-cycle operations. Because such a core is large, the engine has a *single shared* one rather than one per lane: when a warp issues a divide or square root the engine captures all its lanes’ operands, parks the warp on a stall scoreboard, and sequences the active lanes through the shared core one at a time—the same park-and-resume mechanism the memory engine uses, so the scheduler keeps issuing other warps meanwhile.

5.4 Verification

Each unit is verified standalone against a DPI-C IEEE-754 reference: FP32 against the host’s correctly-rounded operations (including the fused `fmaf` and `sqrtof`), and FP16 against an independent double-based reference. Over random vectors spanning the product-dominant, balanced (cancellation), and addend-dominant regimes, plus directed infinity, NaN, zero, flush-to-zero, overflow, divide-by-zero, square-root-of-negative, mis-boxed-operand, and conversion corners, the pipelined unit passes 96,066 checks and the iterative divide/square-root unit a further 20,023, all bit-exact (Table 5.1)—116,089 checks with zero mismatches. End-to-end kernels then run the complete warp pool bit-exact to 1024 threads.

Table 5.2: FP extension area cost: out-of-context synthesis of the accelerator on xczu7ev, integer baseline versus FP-enabled.

Resource	Integer	FP-enabled	Δ
LUTs	30,088	74,342	2.5×
LUTRAM	1,428	3,348	–
Flip-flops	4,590	7,894	1.7×
DSP48E2	24	40	+16
BRAM	0	0	–
Power	0.81 W	1.34 W	–

5.5 Area cost and the timing-closure study

Table 5.2 gives the area cost: floating point raises LUTs from 30,088 to 74,342 (2.5×), flip-flops from 4,590 to 7,894, and DSP blocks from 24 to 40, at 1.34 W. The increase is dominated by the common-case arithmetic unit—eight per-lane copies of the add/multiply/FMA/convert datapath, the FMA being the largest single block—rather than the iterative divide/square-root units, which synthesize to only 689 LUTs each. As a single-cycle block the FMA’s raw path is 16.2 ns (61 MHz), making it the frequency bottleneck of the extension.

The timing closure is a four-step, diagnosis-driven argument, summarised in Table 5.3 and illustrated in Figure 5.1.

Step 1 — two-stage FMA. Pipelining the FP unit into two stages, split at the FMA accumulator boundary, and absorbing the extra cycle in a single-slot warp scoreboard, closes at 78 MHz. The placed critical path is the first FMA stage, but it is now *routing*-dominated (7.4 of 12.8 ns is wire delay): in the FP-enlarged fabric, interconnect, not logic depth, has become the binding constraint.

Step 2 — shared serial SFU. Folding the eight per-lane divide/square-root cores into one shared serial unit removes about 3,700 LUTs and 1,200 flip-flops and drops power from 1.53 to 1.22 W. Crucially it leaves the FMA path structurally unchanged yet, by decongesting the fabric, cuts that path’s routing delay and raises the placed maximum from 78 to 84.1 MHz—same logic, less congestion, higher frequency: a direct confirmation that interconnect bounds the design.

Design-space check. If congestion is the limit, does *trimming* the fabric help? Placing the FP design at four lanes halves its area (37k versus 69k LUTs, 20 versus 40 DSP) but reaches only 79.7 MHz, marginally *below* the eight-lane 84.1 MHz. The critical path is a *single* lane’s FMA cone, whose logic depth is independent of the lane count; the eight-lane configuration therefore Pareto-dominates the four-lane one (equal frequency at

Table 5.3: Placed post-route PPA of the FP-enabled accelerator across the timing-closure steps and the lane design-space point (xczu7ev, out-of-context place-and-route unless noted, 100 MHz constraint). The three-stage FMA (step 3) is what meets 100 MHz; the shared SFU (step 2) and lane trimming help area and power but not the frequency floor. The final row is the complete in-context SoC with ZCU104 pin constraints and a warning-free DRC.

FP datapath	Lanes	LUT	FF	DSP	Power (W)	Fmax (MHz)	Crit. path
Two-stage FMA	8	72,573	12,347	40	1.53	78.0	FMA cone
+ shared serial SFU	8	68,914	11,153	40	1.22	84.1	FMA cone
Three-stage FMA	8	69,987	14,954	40	1.22	100.7	VRF LUTRAM
<i>DSE</i> : four lanes	4	36,785	6,953	20	0.98	79.7	FMA cone
<i>SoC</i> : deployable	8	71,567	17,834	40	1.20	103.7	FP ctl. broadcast

twice the throughput) and the true frequency lever is the FMA *microarchitecture*, not the fabric size.

Step 3 — three-stage FMA. Acting on that finding, the FMA is split across three stages—product, 128-bit align-and-add, and normalize-and-round—so no single stage carries the whole cone. Place-and-route now *meets* timing at 100.7 MHz (worst setup slack +0.071 ns), and decisively the critical path moves *off* the FMA onto the vector-register-file LUTRAM read mux—the same structure that bounds the *integer* design (Section 4.1). The FP datapath is no longer the frequency limiter.

Deployable SoC. Integrated into `chip_top` with the host and the eight-bank shared memory, and after splitting the scheduler’s fetch/execute cloud with a pipeline register, the full chip signs off with every port pin-constrained for the ZCU104 and a warning-free design-rule check at +0.360 ns worst slack (103.7 MHz), 71,567 LUTs, 17,834 flip-flops, 40 DSP, zero BRAM, and 1.20 W: a deployable bitstream, not only a timing report.

5.6 Floating-point throughput

On streaming kernels the added pipeline latency is fully hidden by SIMT warp interleaving; the FP16 points lie exactly on the FP32 curves and the cycle counts are identical, because half precision shares the same pipelined datapath. Half precision therefore delivers reduced storage and bandwidth at *no* throughput penalty. Table 5.4 reports four toolchain-assembled FP application kernels, each verified bit-exact. Two effects stand out: the iterative divide/square-root unit makes `fnorm` about 6× the cycles of the streaming `fsaxpy`, the honest price of correctly-rounded transcendental support on a small iterative core; and `fdot`’s single-lane reduction runs at 45 % lane utilization, so by Eq. (3.1) the idle 55 % is lane-datapath energy a mask-gated design clocks away—an amount that grows because the FP unit makes each lane substantially larger.

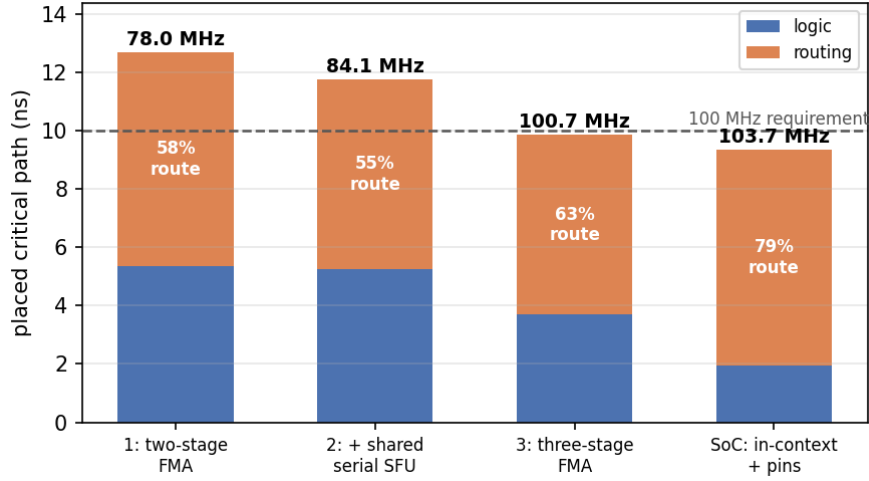


Figure 5.1: Placed critical path at each closure step, split into logic and routing delay. Every step is routing-dominated; the three-stage FMA closes timing by shortening the whole cone, not by reducing the wire share.

Table 5.4: FP application kernels (FP32 unless noted), verified bit-exact. Cycles, global line transactions, and lane utilization at $N=256$.

Kernel	Computation	Cycles	Gmem	Lane-util
fsaxpy	$\alpha A_i + B_i$ (fmadd)	419	96	100%
fsaxpy16	$\alpha A_i + B_i$ (FP16)	451	96	100%
fnorm	$A_i / \sqrt{B_i}$ (div+sqrt)	2,396	96	100%
fdot	$\sum_l A_{8w+l} B_{8w+l}$	1,991	96	45%

Chapter 6

The `pdot8` INT8 Dot-Product Instruction

This chapter begins the transformation of SIMTiX into an AI inference accelerator. The gap between “can run neural-net math” and “is an efficient inference accelerator” is arithmetic precision and density. A general SIMT core *can* do INT8 by masking and shifting bytes, but that costs about eight instructions per four multiply-accumulates (MACs). A single fused dot-product instruction collapses that to one instruction and maps the four multiplies onto the FPGA’s hard DSP48E2 blocks. `pdot8` is that instruction.

6.1 Rationale

A prior resource audit established that SIMTiX uses only about 2.3 % of the device DSP blocks and no BRAM: the FPGA has enormous arithmetic headroom. `pdot8` spends a little of that DSP headroom—four blocks per lane—to buy a 4× INT8 throughput multiplier, exactly the right trade for a design that is nowhere near DSP-bound. Three design decisions define the instruction.

Non-accumulating semantics. `pdot8 rd, rs1, rs2` computes a pure four-way dot product into `rd`; it does not read `rd` as a running accumulator. The integer register file has exactly two asynchronous read ports; a fused `rd += dot` would need a third read port on every lane’s bank. The non-accumulating form keeps the two-port file untouched, and the kernel accumulates in a separate register with a single-cycle add that overlaps the next `pdot8` under the scheduler.

A background DSP engine, not a single-cycle ALU op. `pdot8` executes in a park-and-resume background engine (`W_D0T`), exactly like the RV32M `mul` engine and the FP compute engine, not in the single-cycle ALU. Putting a DSP multiply directly in the single-cycle fetch→execute cloud is precisely what hurt timing historically; a pipelined, parked DSP stays off the critical path, and the scheduler keeps issuing other warps while the DSP tree fills.

Table 6.1: The pdot8 signedness variants (funct3). The signed× unsigned form is the weight×activation case of inference.

funct3	Mnemonic	rs1×rs2	Typical use
000	pdot8	signed × signed	symmetric / weight×weight
001	pdot8u	unsigned × unsigned	activation×activation
010	pdot8su	signed × unsigned	weight × activation (inference)

Custom-0 opcode, encoded with .insn. pdot8 uses the RISC-V custom-0 major opcode, which the ISA reserves for non-standard extensions, so it can never collide with the instructions the toolchain emits and is inert to every existing kernel and to the compiler. Because gcc does not know the mnemonic, kernels encode it with the assembler’s raw `.insn r` directive—no compiler patch is needed.

6.2 Instruction specification

pdot8 is an R-type instruction under the custom-0 opcode. For $rs1 = \{a_3, a_2, a_1, a_0\}$ and $rs2 = \{b_3, b_2, b_1, b_0\}$, where byte i occupies bits $[8i + 7 : 8i]$,

$$rd = \sum_{i=0}^3 \text{ext}(a_i) \times \text{ext}(b_i), \quad (6.1)$$

where $\text{ext}(\cdot)$ is sign-extension for a signed operand and zero-extension for an unsigned one, selected by funct3 (Table 6.1). The sum of four products lies within $[-260100, +260100]$, so it never overflows the 32-bit destination. A single signed 9×9 multiply serves every variant, because an unsigned byte extended with a zero sign bit is non-negative and a signed byte is sign-extended.

6.3 Microarchitecture

pdot8 is a faithful clone of the integer-multiply engine. Operands are captured at issue and held constant while the warp is parked, so the datapath is a fixed-latency streaming pipeline. Per lane, the four byte-pairs are extended to 9-bit signed values (the DSP A/B registers), four signed 9×9 products are formed (the DSP MREG), registered (the DSP PREG), and reduced by a registered adder tree. From issue to a settled result is six cycles: operand capture, A/B register, multiply, product register, and two adder stages. A three-state scoreboard (DOT_IDLE→DOT_RUN→DOT_WB) captures the result one safe cycle after it settles and drives it onto the integer register file when the port is free, then restores the parked warp’s PC. Only one dot is in flight across all warps; a second warp attempting a dot simply waits and retries, which is safe because nothing was committed.

Table 6.2: Measured FPGA PPA of the INT8-enabled accelerator (FP + pdot8), placed out-of-context on xczu7ev at a 100 MHz constraint, versus the FP-only baseline.

Resource	FP + pdot8	FP baseline	Δ
LUT	70,779 (30.7%)	69,987	+792 (~1%)
Flip-flops	17,411	14,954	+2457
DSP48E2	72 (4.2%)	40	+32
BRAM	0	0	0
Placed Fmax	112.2 MHz	100.7 MHz	+11.5
Power	0.969 W	1.22 W	–

Listing 6.1 shows the resulting inner loop.

```
.Lk: lw    av, 0(pa)                # 4 activations (packed int8)
     lw    wv, 0(pw)              # 4 weights    (packed int8)
     .insn r 0x0B, 2, 0, t0, wv, av # pdot8su t0 = sum_i w_i*a_i (4
MACs)
     add   acc, acc, t0            # accumulate (single-cycle, overlaps
)
     addi  pa, pa, 4
     addi  pw, pw, 4
     addi  k, k, -4
     bnez  k, .Lk
```

Listing 6.1: INT8 dot-product inner loop: one pdot8 per four MACs, versus four multiplies and four adds plus byte extraction for a scalar equivalent.

6.4 Verification and cost

The instruction is verified bit-exact: a single warp of eight lanes computes all three variants on distinct operands, including the signed edge values -128 , 127 and $-1/255$ so sign and zero extension are exercised at the boundaries, and every lane’s result is checked against an exact integer reference—all eight lanes and three variants pass. Because the arithmetic is integer, the reference has no tolerance. Table 6.2 gives the measured FPGA cost, placed out-of-context on the xczu7ev against the FP-enabled M17 baseline: the INT8 engine adds 32 DSP blocks (four per lane across eight lanes) and about 1 % of the LUTs, and—crucially—does *not* regress the operating frequency, because it is a pipelined DSP background engine off the fetch→execute path. This is the central design claim of the INT8 extension, confirmed by measurement.

Chapter 7

Quantized GEMM and GEMV Kernels

GEMM (general matrix multiply) is the computational core of every neural-network layer—fully-connected layers *are* GEMMs, and convolution becomes a GEMM after `im2col`—so a correct, coalesced INT8 GEMM is the load-bearing kernel of the inference story. This chapter builds two kernels on `pdot8`: a GEMM tile (`qgemm`) and, for the MLP layers, a GEMV (`qgemv`).

7.1 The INT8 GEMM tile

`qgemm` computes a full 8×8 output tile with reduction length $K = 16$: $C[r][c] = \sum_{k=0}^{K-1} A[r][k] \cdot B[k][c]$, with signed INT8 operands and an INT32 accumulator. One warp maps to each output row and one lane to each output column, so all eight lanes of a warp share the same row of A and each lane owns a different column of B and C . The reduction runs four elements at a time with `pdot8`.

The data layouts are the crux, chosen so that both operand loads coalesce (Table 7.1). A is stored row-major, so all eight lanes of a warp read the *same* address—a single broadcast line transaction. B is stored *pre-packed* as `Bpacked[K/4][N]`, so that for a fixed reduction group the eight columns are contiguous—a single coalesced line. This is the standard inference weight layout: weights are packed offline. The little-endian byte packing matches `pdot8`'s operand format exactly.

7.2 Performance

Table 7.2 gives the measured performance. The kernel does 1024 MACs in 538 cycles, a throughput of 1.90 MAC/cycle against a datapath ceiling of 32 (eight lanes $\times$ four MACs). The 72 global-memory transactions match the analytic coalescing model exactly— $8 \text{ warps} \times (K/4 \text{ loads} \times 2 + 1 \text{ store}) = 8 \times 9$ —an $8 \times$ reduction versus the 576 of a naive per-lane engine, which confirms that coalescing works as designed. The throughput is functional rather than tuned: the gap to the ceiling is two honest and known limits—the single-slot `W_DOT` scoreboard (only one `pdot8` in flight, its latency only partly hidden by eight warps of work), and the per-warp `load`→`pdot8`→`accumulate` dependency chain. The lever for real density is not more SIMT tuning but a tensor/systolic engine, for which

Table 7.1: qgemm data layouts, chosen so both operand loads coalesce to a single line transaction.

Matrix	Layout	Why it coalesces
A	8×16 INT8, row-major	every lane of a warp uses the same row $\Rightarrow$ one broadcast line
B	pre-packed <i>Bpacked</i> [4][8] words	the eight columns of a group are contiguous $\Rightarrow$ one coalesced line
C	8×8 INT32, row-major	the eight columns of a row are contiguous $\Rightarrow$ one store line per warp

Table 7.2: Measured performance of the qgemm INT8 GEMM tile ($M=N=8$, $K=16$).

Metric	Value	Notes
MACs	1,024	$M \cdot N \cdot K = 8 \cdot 8 \cdot 16$
pdot8 SIMT instructions	32	$M \cdot (K/4)$; each spans N lanes $\times 4$
Cycles	538	measured
Throughput	1.90 MAC/cycle	ceiling = 32 (8 lanes $\times 4$)
Global-mem transactions	72	= 8 warps $\times (K/4 \cdot 2 + 1)$

qgemm establishes the correct, coalesced baseline.

At the measured placed power of 0.969 W, and counting operations as twice the MAC rate, the measured efficiency is 0.39 GOP/s/W and the datapath ceiling is 6.6 GOP/s/W at 100 MHz. For an eight-lane academic core these land honestly in the GOPS range, not TOPS; the measured-to-ceiling gap is the motivation for a future tensor engine.

7.3 The GEMV layer primitive

A fully-connected MLP layer with a single input vector is a matrix–vector product (GEMV), not a full GEMM. qgemv computes $out[n] = \sum_k x[k] \cdot W[k][n]$ with one lane per output element n , reducing K four at a time with pdot8. It is parameterized so the *same* kernel runs both MLP layers: the reduction length is read as $K/4$ from a configuration word at a fixed address, and the number of outputs N (the weight-column stride) is the seeded thread count. The input vector x is broadcast (shared by all lanes) and the weights are the same pre-packed *Wpacked*[$K/4$][N] layout as qgemm, so both loads coalesce. Listing 7.1 shows the kernel; it is the primitive the MNIST engine of Chapter 9 launches, once per layer.

```

_start:
    li    t0, 0x100          # config address
    lw    t1, 0(t0)          # t1 = Kp = K/4
    slli  t2, a0, 2          # &Wpacked[0][n] = a2 + n*4
    add   t2, a2, t2
    slli  t3, a4, 2          # N*4 = W column stride (bytes) per kk

```

```

mv      t4, a1          # &x[0]
li      t5, 0           # acc = 0
li      t6, 0           # kk  = 0
.Lk:
lw      s0, 0(t4)        # x[4kk..]          (broadcast)
lw      s1, 0(t2)        # W[4kk..][n]       (coalesced)
.insn r 0x0B, 0, 0, s2, s0, s1 # pdot8 (4 MACs)
add     t5, t5, s2       # acc += partial dot
addi    t4, t4, 4        # x += 1 word
add     t2, t2, t3       # W += N words
addi    t6, t6, 1
blt     t6, t1, .Lk      # uniform loop
slli    s3, a0, 2        # out[n] = acc
add     s3, a3, s3
sw      t5, 0(s3)
ecall

```

Listing 7.1: The parameterized INT8 GEMV kernel. K_p is read from a config word; N is the seeded thread count $a4$. The K-loop is uniform, so it never diverges.

Chapter 8

Requantization

Requantization is the bridge that lets quantized layers be chained. A quantized layer multiplies INT8 activations by INT8 weights into an INT32 accumulator—which is exactly what `qgemm` and `qgemv` produce—but the next INT8 layer needs INT8 *inputs*. Requantization rescales the wide accumulator back to a signed byte, per Eq. (2.1). Without it, the layers cannot compose.

8.1 FP-reuse, not a dedicated datapath

Because SIMTiX already has a full FP32 pipeline (Chapter 5), requantization is a short branchless FP sequence and needs *no new hardware*. This is a deliberate, honest trade. A production FPU-less INT8 accelerator would implement requantize as a fixed-point integer datapath specifically to avoid paying for an FPU; SIMTiX already pays for one, so the marginal-cost-optimal choice is to reuse it. The kernel (Listing 8.1) runs one thread per element: it converts the accumulator to FP32, computes $acc \cdot scale + zp$ with a single fused rounding, clamps in floating point with branchless `fmin/fmax` (so there is no SIMT divergence), rounds to nearest even on the float-to-integer conversion, and stores a signed byte at stride one. Because consecutive lanes’ bytes pack into one word, the output is already in the next layer’s packed-INT8 input layout, with no repacking.

```
_start:                                # a0=tid a1=&Acc(int32) a2=&params(f32) a3
=&Out(int8)
slli    t0, a0, 2
add     t1, a1, t0
lw      t2, 0(t1)                      # acc (int32)
flw     f1, 0(a2)                      # scale
flw     f2, 4(a2)                      # zero_point
fcvt.s.w f3, t2                       # (float)acc
fmadd.s f3, f3, f1, f2                # acc*scale + zp    (fused -> single
rounding)
lui     t3, 0x42fe0; fmv.w.x f4, t3    # 127.0f
lui     t3, 0xc3000; fmv.w.x f5, t3    # -128.0f
fmin.s  f3, f3, f4                    # clamp high
fmax.s  f3, f3, f5                    # clamp low
```

```

fcvt.w.s t4, f3, rne      # round to nearest even -> [-128,127]
add      t5, a3, a0
sb       t4, 0(t5)        # signed byte (packs 4/word)
ecall

```

Listing 8.1: The requantization kernel (FP-reuse). The clamp is done in floating point, before the integer conversion, so it is branchless and does not diverge. A *requant_relu* variant clamps the low bound to 0 to fuse the ReLU activation.

For the MLP an activation-fused variant, *requant_relu*, clamps the low bound to 0 instead of -128 , performing the requantization and the ReLU activation in a single pass—precisely what is needed between two fully-connected layers.

8.2 Precision and verification

The int-to-float conversion is exact for accumulators below 2^{24} ; for the MNIST MLP (784 inputs, $|acc| \lesssim 1.3 \times 10^7 < 2^{24}$) the accumulator is within the exact range, so no precision is lost in the conversion. The kernel is verified bit-exact: eight accumulators spanning the sign range, both clamp saturations, and the boundary values are checked against a software reference, and all eight requantized bytes match. A verification aside worth recording: the first run appeared to fail, but the *reference* was wrong—SystemVerilog’s `int'(real)` cast already rounds to nearest, and the reference had additionally added 0.5, double-rounding one value; the hardware was correct. A golden model is only as trustworthy as its own arithmetic.

Chapter 9

End-to-End On-Chip MNIST Inference

This is the capstone. A trained, quantized $784 \rightarrow 128 \rightarrow 10$ multilayer perceptron classifies MNIST handwritten digits, and the entire forward pass runs end-to-end on the integrated chip: the host CPU streams the weights out of an on-chip block-RAM store into the accelerator’s working memory in tiles, launches the kernels of the previous chapters over the memory-mapped interface, and reads back the predicted digit. It is the proof point that SIMTiX is not a core that merely *can* do neural-network math, but one that runs a real, trained, quantized network end-to-end and gets the right answers.

9.1 The model and its quantization

A $784 \rightarrow 128 \rightarrow 10$ ReLU MLP was trained in pure numpy (SGD, eight epochs), reaching 96.82 % FP32 test accuracy. It was then quantized with symmetric per-tensor INT8: activations and weights scaled to $[-127, 127]$. The INT8 test accuracy is 96.53 %—a drop of only 0.29 %, confirming that INT8 is essentially free in accuracy for this network. The layer-1 requantization multiplier is $M_1 = (S_x S_{w1})/S_h = 0.000696921$. A crucial detail for bit-exact verification: the golden reference computes the requantization in *float32*, matching the hardware FPU exactly rather than in *float64*, so the hardware can be held to a digit-for-digit standard. The model script exports the weights, the test images, and the integer-exact golden predictions as packed hex.

9.2 Per-image inference dataflow

Each 28×28 image is flattened to a 784-vector, quantized to INT8, and run through the network. The forward pass is

$$x(\text{int8}[784]) \xrightarrow{\text{qgemv}} acc_1(\text{int32}[128]) \xrightarrow{\text{requant_relu}} h(\text{int8}[128]) \xrightarrow{\text{qgemv}} acc_2(\text{int32}[10]) \xrightarrow{\text{argmax}}$$

The same `qgemv` runs both layers: for layer 1 the configuration word holds $K/4 = 196$ and the launch requests $N = 128$ outputs; for layer 2 it holds $K/4 = 32$ and $N = 10$. The layouts are chosen so each stage’s output is already the next stage’s input format: `requant_relu` stores the 128 hidden bytes packed four per word, exactly `qgemv`’s packed

Table 9.1: Chip address map for the on-chip MNIST configuration. The block-RAM weight store is a new region (**0xA**); the host CPU stalls one cycle on each weight read to absorb the synchronous-BRAM latency.

Region (top nibble)	Target	Access
0x9	chip result register	write: latch result, raise done
0x8	accelerator MMIO page	command / status
0xA	block-RAM weight store	read-only, 1-cycle stall
else	16 KB LUTRAM working memory	asynchronous, single-cycle

input-vector layout for layer 2, with no repacking.

9.3 The on-chip integration challenge

Running this on the real chip, rather than in a testbench that owns a large flat memory, poses a concrete architectural problem. The accelerator reads its operands from the on-chip shared memory, which is distributed-RAM (LUTRAM) and was enlarged to 16 KB for this demonstration. But the first weight matrix alone, $W_1 \in \mathbb{Z}^{196 \times 128}$ packed, is 25,088 32-bit words—about 98 KB—which does not fit. Placing the full weight set in LUTRAM would cost tens of thousands of LUTs and is impractical. Moreover, the entire chip, host and accelerator alike, assumes an asynchronous single-cycle memory, whereas block RAM is synchronous with a one-cycle read latency; putting BRAM directly in the accelerator’s combinational load path would require risky surgery on the memory engine and would break timing closure.

The chosen architecture leaves the accelerator core *untouched*. The weights live in a dedicated block-RAM store that only the host CPU reads; the CPU *streams* them, in tiles, into the accelerator’s asynchronous LUTRAM working memory. This is exactly the weights-in-BRAM, tile-into-fabric motif that real inference accelerators use. The host CPU already honours a memory-ready handshake (its pipeline stalls while the ready signal is low), so the block RAM’s one-cycle latency is absorbed by a single-cycle CPU stall on each weight read, added in the chip’s top level. Table 9.1 shows the resulting chip address map: a new region routes to the block-RAM weight store, another to the accelerator’s MMIO page, another to the result register, and the remainder to the working memory.

9.4 The host driver

The host driver is a 182-instruction RV32I program, assembled to an instruction ROM. It performs the entire inference under the memory-mapped interface, with no custom host instructions. Because the working memory cannot hold W_1 at once, the driver *tiles* the

first layer: for each tile of 16 output columns (eight tiles cover the 128 outputs) it copies that tile’s weights out of the block-RAM store into the working memory, repacked to the tile’s column stride, and launches `qgemv` on the tile; the second layer’s weights are small enough to stream in a single tile. The driver’s structure per image is:

1. copy the image’s 196 packed words into the working memory;
2. for each of the eight column-tiles of layer 1: stream the tile’s weights (196×16 words, repacked so the kernel reads them at stride 16) from the block-RAM store into the working memory, then launch `qgemv` with $N = 16$ and the output base offset to that tile;
3. launch `requant_relu` over the 128 accumulators to produce the hidden activations;
4. stream W_2 and launch `qgemv` with $N = 10$ for layer 2;
5. argmax the ten logits (a signed integer scan) to obtain the predicted digit, and self-check it against the on-chip golden.

After all 64 images the driver publishes the count of golden-matching predictions to the result register. Two engineering details were essential. First, a *two-phase* kernel-completion poll: the dispatcher parks in its done state with `DONE` held high after each kernel, so a naive poll would immediately observe the *previous* kernel’s stale `DONE`; the driver instead waits first for `BUSY` to assert and only then for `DONE`, which is robust across the roughly 640 kernel launches of the full run. Second, the repacking arithmetic must place tile weights so that the kernel, told a column stride of 16, reads exactly $W_1[kk][n_0 + n]$ —verified against the kernel’s addressing before simulation.

9.5 Result

The design was simulated at the full chip level. The chip boots, the host CPU drives the whole inference, and the result is unambiguous:

golden-match count = 64/64; completed in ~10.7 million cycles.

The chip reproduces the integer-exact software golden for *all 64 test images, bit-for-bit*. Table 9.2 summarises the model and the result. The chip’s predictions match the true image labels on 63 of the 64 images (98.4 %); the single miss is a genuine model misclassification—a sloppily written digit ‘5’ read as ‘6’, not a hardware error, since every one of the 64 outputs is bit-identical to the software golden. Figure 9.1 shows the 64 images the chip classifies, each labelled with the chip’s prediction and the true label.

The existing integer bring-up demonstration (the “result = 964” checksum) shares the same top-level and continues to pass unchanged, so the MNIST configuration is a clean,

Table 9.2: The quantized MNIST MLP and its end-to-end on-chip result.

Quantity	Value
Network	784 $\rightarrow$ 128 $\rightarrow$ 10 MLP, ReLU
FP32 test accuracy	96.82%
INT8 test accuracy	96.53% (−0.29%)
Layer-1 requant multiplier M_1	0.000696921
On-chip result vs. golden	64 / 64 bit-exact
On-chip accuracy vs. labels	63 / 64 = 98.4%
Full-run latency	~10.7M cycles (64 images)
Weight store	256 KB block RAM (host-streamed)

parameter-selected mode of the same chip rather than a fork. This completes the arc from a general SIMT core to a demonstrated, on-chip, end-to-end quantized inference engine.



Figure 9.1: The 64 MNIST test images classified end-to-end on the chip. Each tile shows the exact INT8 pixels the accelerator processes, labelled with the chip’s prediction p and the true label t . The single red label is the one genuine model misclassification (a ‘5’ read as ‘6’); all 64 chip outputs are bit-identical to the software golden.

Chapter 10

Consolidated Results and Discussion

This chapter draws the measured results together and reads them as one story.

10.1 Area and frequency across the design’s evolution

Table 10.1 traces the accelerator through its three major incarnations on the same device. The integer baseline, after the distributed-RAM register file, is a 30,088-LUT, 24-DSP design that meets 100 MHz. The floating-point extension multiplies the LUTs by 2.5× and, through the three-stage FMA closure, still meets 100 MHz with the critical path moved onto the register-file read mux. The INT8 extension then adds 32 DSP blocks and about 1 % of the LUTs on top of the FP design without regressing frequency (112.2 MHz placed). The consistent thread is that the arithmetic extensions are cheap in LUTs and are kept off the critical path; the binding constraint throughout is the register-file read multiplexer and interconnect congestion, not the compute.

10.2 Reading each result

The LUTRAM result (Table 4.1). The 5.7× LUT reduction is the enabling result of the whole thesis: it is what makes a readable SIMT design also a *fittable* one, and it identifies the register file, not the ALUs, as the true cost centre—a lesson that recurs at every later stage, where the register-file read mux keeps re-emerging as the critical path.

The benchmark speedups (Table 4.3). The 7–10× on streaming kernels, 2.76× under heavy divergence, and 0.72× on the serial reduction together map the SIMT model’s operating envelope honestly. The sub-unity reduction is not hidden; it says precisely that a reduction needs a dedicated hardware primitive, and it is consistent with the 39.8 % lane utilization that Eq. (3.1) converts into a large clock-gating opportunity.

The FP verification and closure (Tables 5.1, 5.3). The 116,089 bit-exact checks establish IEEE-754 correctness including the fused multiply-add and iterative divide/square root in both precisions. The closure table is a diagnosis in five rows: the two-stage FMA exposes a routing-bound path, the shared SFU confirms congestion by *raising* frequency

Table 10.1: The accelerator across its three incarnations (placed out-of-context on `xczu7ev`). Each extension is deliberately kept off the frequency-limiting path.

Configuration	LUT	FF	DSP	Placed Fmax
Integer (LUTRAM VRF)	30,088	4,590	24	≥ 100 MHz
+ Floating point (3-stage FMA)	69,987	14,954	40	100.7 MHz
+ INT8 <code>pdot8</code>	70,779	17,411	72	112.2 MHz

with unchanged logic, the four-lane point rules out fabric size as the lever, and the three-stage FMA closes 100 MHz by shortening the cone itself—after which the deployable SoC signs off at 103.7 MHz.

The INT8 cost (Table 6.2). The design claim—INT8 throughput for 32 DSP and 1 % LUT, with no frequency regression—is confirmed by measurement, and it is confirmed *because* the engine is a pipelined DSP kept off the fetch→execute path.

The GEMM throughput (Table 7.2). The $8\times$ memory-traffic reduction matches the analytic coalescing model exactly, validating the layout design; the 1.90-of-32 MAC/cycle throughput is reported with its two known causes and its clear next lever (a tensor engine).

The MNIST result (Table 9.2). The 64/64 bit-exact on-chip result is the thesis’s proof point: a real, trained, quantized network runs end-to-end through the integrated chip and matches the software golden digit for digit, with the weights streamed from block RAM exactly as a deployable inference accelerator would.

10.3 Limitations and honest boundaries

Several results are deliberately reported as limits rather than smoothed over. The single shared data port caps SIMT throughput below the lane-count ideal. The serial reduction is slower than the scalar host. The `qgemm` throughput is far below its datapath ceiling because of the single-slot dot engine and the per-warp dependency chain. The lane clock-gating saving is an analytic model from hardware counters, not a gate-level power measurement. And the block-RAM-inclusive PPA of the MNIST chip configuration—as opposed to the accelerator-only INT8 PPA of Table 6.2—is the one measurement not yet taken; it is the immediate next step, and it is expected to show the first non-zero block-RAM usage in the design (roughly 57 block-RAM tiles for the 256 KB weight store). Reporting these boundaries is part of the contribution: the evaluation is a map of where the design does and does not pay, not a highlight reel.

Chapter 11

Conclusion and Future Work

11.1 Conclusion

This thesis has presented SIMTiX, a compact, fully-open RISC-V SIMT accelerator, and carried it from a general-purpose parallel core to a demonstrated, on-chip, end-to-end quantized neural-network inference engine. The baseline design fits the complete SIMT mechanism set—warp scheduling and latency hiding, memory coalescing, a reconvergence stack, and a scratchpad—into a readable, synthesizable form that runs through the entire flow to a routed bitstream, and a distributed-RAM register file is the key that makes it small ($5.7\times$ fewer LUTs) and fast (100 MHz) on FPGA. A verified six-kernel suite gives a measured $7\text{--}10\times$ speedup where the SIMT model fits, degrades gracefully under divergence, and—reported honestly—loses on a serial reduction. A per-lane floating-point unit adds FP32 and FP16 arithmetic, with half precision obtained almost for free from the single-precision path and verified bit-exact over 116,089 vectors, and a diagnosis-driven timing study takes the FP-enlarged design to a deployable 103.7 MHz SoC. Finally, a custom INT8 dot-product instruction, quantized GEMM/GEMV and requantization kernels, and a block-RAM weight-streaming architecture compose into a trained, quantized MNIST classifier that runs end-to-end on the integrated chip and reproduces the integer-exact golden on all 64 test images, bit-for-bit. The design and its evaluation infrastructure are public, so the results are reproducible and the platform is reusable for teaching and research.

11.2 Future work

The evaluation points to concrete next steps.

- **A tensor / systolic engine.** The measured-to-ceiling gap in `qgemm` is the clearest signal that real INT8 density needs an output-stationary MAC array, which is also where the “two INT8 per DSP” weight-sharing packing applies. This is the natural successor to `pdot8`.
- **Block-RAM-inclusive PPA of the MNIST chip.** The accelerator-only INT8 PPA is measured; synthesizing the full MNIST chip configuration would report the first

block-RAM usage and complete the deployable-inference PPA picture.

- **A dedicated cross-lane reduction primitive**, to convert the reported reduce loss into a win, and **multiple or banked memory ports**, to raise the throughput saturation point above the single-port limit.
- **A dedicated integer requantization instruction**, to enable an FPU-less INT8 inference path and a direct area/energy comparison against the FP-reuse kernel.
- **An open-source ASIC path** through an open PDK, to turn the FPGA PPA numbers into silicon-relevant area, frequency, and power.

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